PHYSIQUE & PERFORMANCE OPTIMIZATION · TRT + HCG + SERMORELIN (PERFORMANCE / QUALITY-OF-LIFE PROTOCOL) · SLOW-TWITCH · DOB MAY 30 1975 · V14 · AUGUST 15 2026

Ken Mark — Physique & Performance Dashboard

● TRT Start: Aug 2024 ● Serious Training: Nov 28 2025 ● Sermorelin: Dec 22 2025 ● Anastrozole ↻ Restarted May 13 2026 · 0.25mg × 2/wk ● T Dose ↑ 100mg/wk: Apr 19 2026 ● Bloodwork: Jul 28 2026

● Hamstring strain · 4 tweaks Jun 19 – Jul 19 2026 ● Garmin current to: Jul 21 2026 ● **Latest eVolt: May 18 2026 · Age 51 (Guelph)**

GOAL HIERARCHY (V12) **Primary: physique** (muscle, body composition). **Secondary but genuinely important: performance** — anchored to VO2 max upkeep and a personal connection to running; not a co-equal headline goal. **The real performance targets are absolute VO2 (~3,571 ml/min) and interval quality, not relative VO2** — relative decline (47.5 → ~42.7 by 2030) is a denominator effect from added mass, not a regression. The hormone/training protocol is run for these ends and for recovery and quality of life — **it is not framed as a longevity or anti-aging intervention.** **Secondary / monitored separately:** longevity is de-emphasized here and tracked as a follow-up to-do list of high-evidence markers not yet in the dataset (ApoB, Lp(a), blood pressure, sleep, CAC, screening) — see *Appendix · Longevity Watch-List* at the end. Where a protocol lever trades long-term considerations for performance/QoL, that trade is named there, eyes open.

CURRENT STATUS

MUSCLE GAINED · ADJUSTED	CURRENT MUSCLE · ADJUSTED	CURRENT BODY FAT · ADJUSTED	PROJECTED END 2030	VO2 MAX · PERFORMANCE
+9.3	78.3	16.0%	88.3	47.0
lbs since Aug 2024 · headline +13.2	lbs · May 18 · headline 82.2	headline 11.4% · water-corrected	lbs muscle · age 55 · physique target	ml/kg/min · Garmin Jul 2026 · peaked 48 May–Jun

BODY COMPOSITION & VO2 MAX PROJECTION

YEAR / AGE	WEIGHT (LBS)	MUSCLE (LBS)	FAT (LBS)	BF%	VO2 MAX	VO2 TIER	MUSCLE Δ YOY / TOTAL VS BASELINE
2024 Age 49 AUG 2024	155.0	69.0	23.2	15.0%	47.5	EXCELLENT	—
2025 Age 50 END 2025	165.0	77.9	24.1	14.6%	46.2	EXCELLENT	—
2026 Age 50 FEB 2026	167.5	79.8	24.5	14.6%	47.0	EXCELLENT	—
2026 Age 50 MAY 18 2026 ⚠ OUTLIER	165.8	82.2	19.0	11.5%	47.5	EXCELLENT	—
2026 Age 51 AUG 15 2026 ◀ ANCHOR	166.7	78.3	26.7	16.0%	47.2	EXCELLENT	—
2026 Age 51 END 2026 ▶	168.2	79.3	26.1	15.5%	46.8	EXCELLENT	+1.0 lbs (Aug-Dec 2026 (~4.5 mo)) / +10.3 total vs baseline
2027 Age 52	169.7	82.3	24.6	14.5%	46.4	EXCELLENT	+3.0 lbs (Full year) / +13.3 total vs baseline
2028 Age 53	171.6	84.8	24.0	14.0%	45.9	EXCELLENT	+2.5 lbs (Full year) / +15.8 total vs baseline
2029 Age 54	172.9	86.8	23.3	13.5%	45.5	EXCELLENT	+2.0 lbs (Full year) / +17.8 total vs baseline
2030 Age 55	173.7	88.3	22.6	13.0%	45.3	EXCELLENT	+1.5 lbs (Full year) / +19.3 total vs baseline

BLOODWORK HISTORY — KEY MARKERS (8 DRAWS)

DATE & CONTEXT	TOTAL T NMOL/L	FREE T PMOL/L	ESTRADIOL PMOL/L	HEMATOCRIT	HGB G/L	TSH MIU/L	CORTISOL AM NMOL/L	CRP MG/L
Aug 1 2024 Pre-TRT baseline	22.1 8.4–28.8	334 196–636	97 <162	0.442 0.40–0.50	147 135–175	2.35 0.32–4.00	97 140–535	<0.5 <5.0
Oct 23 2024 TRT ~2 months	31.1 8.4–28.8	523 196–636	— <162	0.466 0.40–0.50	155 135–175	— 0.32–4.00	— 140–535	— <5.0
Feb 3 2025 Dose too high · supraphysiol.	44.8 8.4–28.8	908 196–636	46 <162	0.499 0.40–0.50	169 135–175	3.4 0.32–4.00	49 140–535	7.2 <5.0
Feb 10 2025 Above lab max · dose crisis	>52 8.4–28.8	n/calc. 196–636	— <162	0.511 0.40–0.50	171 135–175	— 0.32–4.00	— 140–535	— <5.0
May 13 2025 Adjusted · hct still elevated	30.6 8.4–28.8	620 196–636	<40 <162	0.529 0.40–0.50	175 135–175	2.88 0.32–4.00	253† 140–535	<0.5 <5.0
Feb 24 2026 Well-managed · TSH elevated	28.1 8.4–28.8	491 196–636	<40 <162	0.522 0.40–0.50	171 135–175	4.74 0.32–4.00	249 140–535	<0.5 <5.0
May 12 2026 Post-AI · dose ↑ 100mg/wk	32.7 8.4–28.8	667 179–475 *	194 <162	0.498 0.40–0.50	171 135–175	4.34 0.32–4.00	270 140–535	<0.5 <5.0
Jul 28 2026 Post-illness ‡ · E2 on target	33.9 8.4–28.8	784 179–475 *	113 <162	0.521 0.40–0.50	172 135–175	3.86 0.32–4.00	292 140–535	2.5 <5.0

** May 12 / Jul 28 2026 Free T reference is 179–475 pmol/L (LifeLabs). Prior draws referenced 196–636 pmol/L (different lab). † Random draw, not AM-fasted. ‡ Jul 28 draw taken 1 day after a cold (Jul 22–27) resolved and after a day of low fluid intake — CRP 2.5 and ferritin 174 are both acute-phase-confounded. **Jul 28 additions not shown above:** IGF-1 134 · Vit D 57.1 LO · PSA 1.6 · LDL 2.55 · HDL 1.58 · TG 0.47 · ApoB not ordered. **Still not on panel:** ApoB, Lp(a), cystatin C — queued in Appendix.*

⚠ MODEL RE-ANCHOR (V13) · MAY 18 RECLASSIFIED AS OUTLIER · PROJECTION REBASED

eVolt derives skeletal muscle from total body water, so scans are only comparable at matched hydration. Four scans now exist; two at Optimal TBW, two at High:

Oct 15 2025	TBW 100.1 *Optimal*	muscle 77.6	BF 14.3%	visceral 80	waist 31.2"
Feb 19 2026	TBW 103.0 **HIGH**	muscle 79.8	BF 14.6%	visceral 81	waist 31.6"
May 18 2026	TBW 105.8 **HIGH**	muscle 82.2	BF 11.5% *Under*	visceral 59	waist 30.2"
Aug 15 2026	TBW 100.8 *Optimal*	muscle 78.3	BF 16.0%	visceral 88	waist 32.1"

The hydration-matched pair is Oct 2025 → Aug 2026 — 0.7 lbs apart on water, 10 months apart in time. Muscle 77.6 → 78.3 = **+0.7 lbs, inside BIA test-retest noise (±0.8–1.6 lbs at this mass).** True change: flat to slightly positive. *No meaningful muscle has been lost.*

May 18 is now a documented outlier, not a data point. Highest water, lowest fat mass (flagged Under), lowest visceral area, smallest waist — every metric moved favourably at once and then reverted. E2 was 194 pmol/L six days prior; fluid retention is the parsimonious explanation. The v11/v12 "water correction" of 82.2 → 82.0 was far too small.

Consequence: the End-2026 target of 86.8 lbs is unreachable (it would need +8.5 lbs in 4.5 months) and the 2030 target of 101.3 lbs was built on the same inflated base. Projection rebased from the Aug 15 anchor at rates consistent with observed progress. New End-2030 figure: **88.3 lbs**. This is a substantive model change made at Ken's instruction on Aug 15 2026.

◇ REAL SIGNAL · MODEST FAT ACCUMULATION OVER 6 MONTHS

Comparing like to like (Feb High-water → Aug Optimal is imperfect; Oct Optimal → Aug Optimal is clean): **waist 31.2 → 32.1 in (+0.9)**, **visceral area 80 → 88 cm² (+10%)**, **body fat 14.3% → 16.0%** over 10 months. Abdominal circumference is a tape measurement and is hydration-independent — this part is real, if small. Consistent with the three-companies confounder: compromised sleep preferentially drives visceral adiposity, and there has been no running since late July. Prior v12-session claims of "+1.9 in waist" and "+49% visceral" were artifacts of using May as the baseline — corrected here.

★ HEADLINE · ESTRADIOL 194 → 113 PMOL/L — LANDED MID-TARGET

The May 13 half-dose Anastrozole restart (0.25mg × 2/wk) worked exactly as intended. E2 now sits inside the 70–150 target band, below the lab ceiling of 162. **Do not add DIM, calcium-D-glucarate, chrysin, nettle root or high-dose zinc** — all have aromatase-inhibiting activity and would stack on top of the prescribed AI. Lower is not better here: see the connective-tissue alert below.

⚠ NEW · VITAMIN D 57.1 LO — CONNECTIVE-TISSUE RISK STACK

Below the lab deficiency floor of 75.0 nmol/L. Three factors now converge on bone and tendon quality simultaneously: **vitamin D 57.1 (deficient)**, **calcium 2.15 (exact bottom of range)**, and **E2 held at 113 by prescribed Anastrozole** — estradiol is the dominant regulator of male bone density and associates with tendon quality. This overlaps the hamstring window (4 tweaks, Jun 19 – Jul 19) and is a plausible third strand alongside the speed-exposure gap and sleep restriction. **Dr Salim (Jul 30): D3 2,000 IU daily — NOT 4,000** (hypercalcemia / kidney-stone risk). Take with a fat-containing meal. Retest at 3 months.

✓ CLEAN · LIPIDS · PSA · RENAL · GLYCEMIC

LDL 2.55 · HDL 1.58 · TG 0.47 · Chol/HDL 2.8 — all excellent. PSA 1.6 (first draw, clean baseline). Creatinine 93 / eGFR 86 — **creatine 5g/day inflates serum creatinine independently of muscle mass; cystatin C requested for a clean renal read.** HbA1c 5.4 → 5.5, glucose 5.2 → 5.4 — small, but sleep restriction impairs insulin sensitivity, so no longer filed as pure noise. Prolactin 19.4 HI → 17.8 (resolved). ALT 33 → 22. TSH 4.34 HI → 3.86 (in range).

★ HEADLINE · MAY 18 EVOLT — RECALIBRATED (SMALL WATER CORRECTION)

Adjusted: +2.2 lbs muscle / −3.5 lbs fat / BF 14.6 → 12.7% over 13 weeks. (eVolt headline: +2.4 / −5.5 / 11.4%.) Initial water-artifact correction was overstated — Feb 19 TBW was already High at 46.7 kg, so the differential to May (ECF +0.3 kg) is small. Most of the May 18 changes are real. Visceral fat level 9 → 7 and visceral fat area 81 → 59 cm² are independent confirmation. Pace on track for End 2026 muscle target (86.8 lbs).

↻ IN MOTION · ANASTROZOLE RESTARTED MAY 13 — LOW DOSE (0.25MG × 2/WK)

In response to May 12 draw showing E2 at **194 pmol/L** (30% above target band, above lab ceiling 162), Dr Salim restarted Anastrozole at **half the prior dose**: 0.25mg twice weekly (0.5mg/wk total). 5 days in as of May 18. Anastrozole half-life ~50 hrs — steady state ~3-4 wks. Expected trajectory: E2 settling into 70-150 pmol/L target band. Watch for symptoms in either direction (water retention/mood lability = still high · joint dryness/low libido = over-corrected). Note: elevated E2 may have contributed to high Total Body Water (ECF 16.2 kg) on May 18 scan — expect plasma volume to normalize as E2 drops. Retest at next routine draw (~6-8 wks).

↑ MONITOR · HEMATOCRIT 0.498 → 0.521 — REAL RED-CELL GAIN · DR SALIM THRESHOLD 0.53

RBC 5.26 → 5.53 (+5.1%) with Hgb flat at 172 g/L. Since Hct = RBC × MCV and MCV fell 95 → 94, the entire rise is *cell count, not concentration*. This rules out both hemoconcentration (which would raise Hgb proportionally — Ken reported low fluid intake the prior day) and sample artifact. Contrast with the May draw, where Hgb held flat while Hct *fell* — that was plasma volume expansion; this is the opposite mechanism. **Dr Salim (Jul 30): no donation required below 0.53 — continue monitoring; no room to increase T dose at this time.** Dashboard threshold corrected from 0.50 to 0.53 to match treating physician. Secondary signal: MCH 32.5 → 31.1, iron 21.9 → 17.3, TSAT 0.35 → 0.30 — possible early iron-restricted erythropoiesis (marrow outrunning iron supply). Weak: RDW unchanged at 11.7, which usually widens if true. Ferritin 108 → 174 is acute-phase-confounded (CRP 2.5). Reticulocyte count requested to test this.

↓ MONITOR · TSH 4.34 H (FREE T4/T3 NORMAL)

TSH still elevated but trending down from 4.74. **Free T4 13** (mid-range, 9-19) and **Free T3 4.7** (mid-range, 2.6-5.8) rule out true hypothyroidism — confirms Dr Salim Mar 12 read that TSH elevation is mild GH-axis artifact from Sermorelin. No action required; re-check next draw.

↓↓ ESCALATED · IGF-1 214 → 183 → 134 UG/L — THIRD CONSECUTIVE DECLINE

-37% since Feb, -27% in the last 11 weeks. Now below the 161 pre-Sermorelin baseline attributed to Dr Salim (Mar 12) — *note: that 161 figure is carried from a prior session and the source lab report has not been located. Treat as unconfirmed.* Value remains mid-range for the assay (66-225) and is unremarkable for a 51-year-old male; this is a protocol-efficacy question, not a health finding.

Differential, ranked: (1) **Sleep restriction** — most GH secretion occurs during slow-wave sleep; three companies founded from late March (see Context Block) with sleep compromised over exactly the Feb→Jul window. Tightest temporal fit. (2) **Acute illness** — cold Jul 22-27, draw Jul 28, CRP 2.5; illness suppresses the GH axis transiently. (3) **Peptide handling** — several weeks overseas travel; Sermorelin degrades without consistent refrigeration. Free to check. (4) **Tachyphylaxis** — receptor downregulation from 5-day/wk dosing without washout.

Prior v11 read that TSH normalization corroborates GH-axis decline is now weakened: ashwagandha (undisclosed until Aug 2026) has documented thyroid effects and is a competing explanation for TSH 4.74 → 4.34 → 3.86. Free T4/T3 requested. **Action:** retest IGF-1 when healthy and sleeping normally, recording hours since last injection. Dr Salim did not comment on this item — open question.

↑ NOTE · FREE T 667 H · TOTAL T 32.7 H

Up from 491 / 28.1 — consistent with the 80→100mg/wk dose increase and stopped Anastrozole (less aromatization sink, more T available). Total T modestly above range (28.8 ceiling); Free T 40% above LifeLabs range (475 ceiling). Dr Salim's stated goal was "higher but safe range" for mood/libido. Steady state ~6 wks from dose change — next draw is the true read.

✓ METABOLIC WINS · HBA1C 5.4 · GLUCOSE 5.2 · CRP <0.5

HbA1c dropped 5.6 → **5.4%** — now safely below 5.5% at-risk threshold. Fasting glucose 5.5 → 5.2. CRP remains <0.5 (excellent, no cardiovascular inflammation). Creatinine 91 (up from 80); eGFR 89 well above 60 threshold — the drop is an artifact, not kidney function decline. **Attribution corrected (v11):** added muscle mass is *not* the sole cause. Creatine monohydrate supplementation (5-10g daily) independently raises serum creatinine by roughly 10-25 μmol/L through direct conversion, entirely independent of muscle accrual or renal function. Both factors are in play here and cannot be separated from this panel alone. Practical consequence for Dr Salim: eGFR is unreliable as a kidney-function read while creatine is being taken — **cystatin C** is the marker that bypasses the artifact if renal function ever needs a clean assessment. Prolactin 19.4 marginal (lab note: repeat).

PURPOSE FRAMING · READ BEFORE THE TABLE

This protocol is run for **physique, performance, recovery, and quality of life** — not as a longevity intervention. Three levers in particular are **performance/QoL choices, not anti-aging ones**, and a couple run against the longevity literature: (1) **Sermorelin raises GH/IGF-1** for recovery and fat oxidation, whereas *lower* GH/IGF-1 tone is what tracks with exceptional human longevity and lower cancer risk (Milman 2014; Junnila 2013) — so wanting IGF-1 *higher* is a physique call, not a lifespan one; (2) **free T pushed above range** ("higher but safe") is for mood/libido/drive, with no longevity rationale; (3) **hematocrit** is the one near-term hard-endpoint risk in the stack and deserves a pre-set action threshold. None of this argues for stopping — it argues for choosing each lever with eyes open. Trade-offs catalogued in *Appendix · Longevity Watch-List*.

INTERVENTION	STARTED	DOSE & FREQUENCY	PURPOSE	PROJECTION / STATUS IMPACT
Testosterone Enanthate Hikma Canada · 200mg/ml	Aug 2024 ↑ **Apr 19 2026**	**100mg/wk** · 50mg IM Day 1 + 50mg IM Day 4 (0.25ml × 2) Previously: 40mg × 2 = 80mg/wk	Androgen replacement · anabolic signaling · MPS Above-range free T = performance / mood / libido (QoL) choice, not longevity	Total T 33.9 / Free T 784 (Jul 28) confirm dose response. **Dose held — no increase while Hct elevated (Dr Salim Jul 30).** Monitor Hct (now 0.498) & E2 (now 194) at next draw.
HCG	Aug 2024	Per Dr Salim	HPG axis preservation · intratesticular T · fertility	FSH <0.2 / LH <0.1 — exogenous T suppressing axis as expected. Tissue quality preserved by HCG.
Anastrozole RESTARTED · low dose	Aug 2024 → Stopped Apr 18 2026 ↻ **Restarted May 13 2026**	**0.25mg × 2 / wk · 0.5mg total** Half of prior dose · resumed in response to E2 194 (May 12 draw)	Aromatase inhibition · E2 control	Restart matches the projected path from the May 12 alert. Steady state ~3-4 wks. Target E2 70-150 pmol/L. Retest at next routine draw (~6-8 wks).
Sermorelin GHRH analogue	**Dec 22 2025**	0.3ml SC · 5 days/week (weekends off — receptor sensitivity)	Endogenous GH stimulation · IGF-1 · recovery · fat oxidation Performance / recovery lever — explicitly NOT an anti-aging use (see Appendix)	IGF-1 214 → 183 (still well above 161 pre-Sermorelin baseline). TSH 4.34 confirms mild GH-axis artifact persists; Free T4/T3 mid-range normal.
Heavy Weight Training Sporadic 1x/wk: Nov 2024-Nov 2025	**Nov 28 2025**	3-4 sets to failure · 8-10 reps · 2 days rest/muscle group · 6-day rolling cycle	Mechanical overload · progressive hypertrophy	~24 weeks of serious training as of May 2026. Run Phase 1 active: 1 Helgerud 4×4 + 1 Tempo per cycle.

PHYSICIAN DECISION LOG — NEW IN V12

✓ DR SALIM · EMAIL JUL 30 2026 — FOUR DECISIONS

1. Hematocrit threshold is 0.53, not 0.50. Blood donation not required unless Hct reaches 0.53 or above; continue close monitoring. *Dashboard corrected — the prior 0.50 target was carried from project instructions and generated false alarms.*
2. Vitamin D3 2,000 IU daily, not 4,000. Very high vitamin D can raise calcium and increase kidney-stone risk.
3. Ashwagandha: can slightly raise testosterone on bloodwork — stop several days before any future draw.
4. Testosterone dose held. No room to increase while Hct is elevated. Continue current 100mg/wk.
5. Repeat bloodwork in 3-4 weeks — approved.

◇ OPEN ITEMS — NOT ADDRESSED BY DR SALIM

(a) Sleep apnea screening — no response; Ken elected to skip for now (see Life Context).
(b) IGF-1 134 and whether Sermorelin remains worth 5 injections/week — no response. Ongoing cost and effort with an unresolved efficacy question.
(c) Requisition contents unconfirmed — Ken to confirm ApoB, Lp(a), cystatin C, reticulocyte count, free T4/T3 and vitamin D are all on the 3-4 week draw, or the panel returns a repeat CBC and little else.
(d) Vitamin C — no comment; Ken has stopped it.

EVOLT SCAN COMPARISON — 3 SCANS · RECALIBRATED MAY 18 WATER CORRECTION

METRIC	OCT 15 2025	FEB 19 2026	MAY 18 (EVOLT)	MAY 18 ADJ ◄	Δ FEB → MAY (ADJ)	NOTES
Total Weight	73.5 kg · 162.0 lbs	76.0 kg · 167.5 lbs	75.2 kg · 165.8 lbs	**75.2 kg · 165.8 lbs**	−0.8 kg / −1.7 lbs	Scale measurement · no adjustment
Skeletal Muscle	35.2 kg · 77.6 lbs	36.2 kg · 79.8 lbs	37.3 kg · 82.2 lbs	**37.2 kg · 82.0 lbs**	+1.0 kg / +2.2 lbs	Modest −0.2 lb water correction · headline +2.4 lbs
Fat Mass	10.5 kg · 23.1 lbs	11.1 kg · 24.5 lbs	8.6 kg · 19.0 lbs	**9.5 kg · 21.0 lbs**	−1.6 kg / −3.5 lbs	Closer to headline than v1 correction suggested · headline −5.5
Body Fat %	14.3% (Under)	14.6% (Under)	11.4%	**12.7%**	−1.9 pts	Revised correction: +1.3 pts (was +1.5) · headline 11.4% / −3.2 pts
Lean Body Mass	63.0 kg	64.9 kg	66.6 kg	**66.2 kg**	+1.3 kg	Includes muscle, water, bone
Total Body Water	45.4 kg (Optimal)	46.7 kg (High)		**48.0 kg (High)**	+1.3 kg vs Feb	Oct = baseline · Feb already elevated · May incremental rise small
Intracellular Fluid	30.3 kg	30.8 kg		**31.8 kg**	+1.0 kg vs Feb	ICF in muscle cells = real hypertrophy signal · genuine gain
Extracellular Fluid	15.1 kg	15.9 kg		**16.2 kg**	+0.3 kg vs Feb	Small ECF rise = small water artifact · basis for revised correction
BMR	1,730 kcal	1,771 kcal		**1,808 kcal**	+37 kcal	Reflects added muscle · TEE 2,784 kcal
Visceral Fat Level	9 (Balanced)	9 (Balanced)		**7 (Balanced)**	−2 levels	Not water-affected · independent confirmation of fat loss
Visceral Fat Area	80 cm² (Optimal)	81 cm² (Optimal)		**59 cm² (Optimal)**	−22 cm²	Substantial visceral fat reduction Feb → May
Bio Age	44	45		**43 (eVolt)**	−2 yrs (eVolt)	eVolt = body-comp algorithm only. A separate "fitness-age proxy" (~30, range 28-33) is a bespoke, unvalidated composite — de-emphasized in v10; see Appendix.
Protein	13.3 kg (High)	13.7 kg (High)		**14.2 kg (High)**	+0.5 kg	Continuing to build · less water-sensitive
Grip Strength	—	—		**R 61.3 kg · L 58.9 kg (Handeful)**	new metric	~49 kg Jamar-equivalent (−20%) · ≥90th percentile age 51 · R/L ratio 1.04
BWI Score	8.3	8		**8.6**	+0.6	eVolt composite wellness index
Scan Location	GoodLife #281 Eramosa Rd	GoodLife #242 Pergola		**GoodLife #242 Pergola (same as Feb)**	—	Feb→May on same device · Oct on different device · ±0.5-1 kg lean variance

Recalibrated adjustment methodology: *Feb 19 TBW was already High at 46.7 kg (ECF 15.9). May 18 TBW 48.0 (ECF 16.2). ECF differential is only +0.3 kg — small. Revised correction (vs prior over-correction of −0.5 muscle / +1.5 pts BF):* ***−0.2 lbs muscle, +1.3 pts BF****. Cross-check: at 165.8 lbs / 12.7% BF, fat = 21.0 lbs / lean = 144.8 lbs / muscle (lean − OTHER 62.8) = 82.0 lbs. Internally consistent.* ***Device variance caveat:*** *Feb→May on same device (Pergola #242); Oct on different device (Eramosa #281). eVolt-to-eVolt comparisons across machines can carry ±0.5-1 kg lean variance — Oct→Feb deltas modestly less precise than Feb→May. Segmental (May 18): R Leg Lean 10.05 kg · L Leg Lean 9.78 kg · R Arm 3.73 kg · L Arm 3.75 kg · Torso 28.35 kg.*

MODEL NOTES & ASSUMPTIONS

PROJECTION METHODOLOGY & MAY 2026 TAKEAWAYS

Baseline (Aug 2024): 155 lbs · ~15% BF · muscle ~69.0 lbs (back-calculated). Pre-TRT, pre-training. All "total vs baseline" figures calculated from this point.

Training phases: Nov 2024–Nov 2025 = sporadic ~1×/week, half required volume. Serious training (3-4 sets to failure, 8-10 reps, 2 days rest/muscle) began Nov 28 2025 only. ~24 weeks of serious training as of May 18 2026.

May 18 2026 = new current anchor (recalibrated water correction). eVolt headline 165.8 lbs · 82.2 muscle · 19.0 fat · 11.4% BF. **Adjusted: 82.0 muscle · 21.0 fat · 12.7% BF.** Weight unchanged (scale read). Earlier correction (−0.5 muscle / +1.5 BF pts) was over-aggressive; recalibrated to −0.2 muscle / +1.3 BF pts based on actual ECF differential between scans.

Recalibration logic: Feb 19 TBW was already High at 46.7 kg (this only became visible when full Feb scan data was retrieved). ECF rose only +0.3 kg from Feb to May (15.9 → 16.2 kg). Differential water inflation in lean compartment is small. ICF rose +1.0 kg over the same period — that's water inside muscle cells, a real hypertrophy signal, not artifact.

Adjustment methodology: muscle −0.2 lbs, BF% +1.3 pts vs headline. Cross-check: at 165.8 lbs / 12.7% BF, fat = 21.0 lbs / lean = 144.8 lbs / muscle (lean − OTHER 62.8) = 82.0 lbs. Internally consistent.

Feb → May pace check (adjusted): +2.2 lbs muscle / −3.5 lbs fat over 13 wks. Muscle rate ~0.17 lbs/wk — essentially matching the prior 0.18 lbs/wk eVolt-implied pace. Earlier "decelerated to 0.15" claim was an artifact of the over-correction.

End 2026 muscle target retained at 86.8 lbs: Adjusted May position (82.0) needs +4.8 over ~7.5 months = ~0.6 lbs/month. Current adjusted pace ~0.7 lbs/month — on track or slightly ahead.

BF% trajectory revised: Adjusted current 12.7%. End-2026 → 2030 glide path: 12.0 / 11.5 / 11.0 / 11.0 / 11.0%. Realistic on the adjusted scale; expect future eVolt headlines to read 0.5-1.5 pts lower when TBW is elevated.

Total muscle gained since Aug 2024 baseline (adjusted): +13.0 lbs (headline +13.2). Smaller correction than the prior +12.7 figure. Still well within upper range of expected response for 24 wks of serious training on a four-pillar protocol.

Device variance caveat: Feb 19 and May 18 scans were on the same machine (GoodLife #242 Pergola). Oct 15 was on a different machine (GoodLife #281 Eramosa). Cross-machine eVolt comparisons can carry ±0.5-1 kg lean variance — Oct→Feb deltas are slightly less precise than Feb→May. The TBW rise (45.4 → 46.7 kg) is a real fluid-state change, but device offset means small numerical deltas warrant interpretive caution.

2027–2030 muscle: +5.0 / +4.0 / +3.0 / +2.5 lbs/yr. Target ~101.3 lbs muscle by end 2030 (age 55), +32.3 lbs vs Aug 2024 baseline.

OTHER constant note: OTHER = 62.8 lbs held as modeling anchor. May 18 adjusted residual = 165.8 − 82.0 − 21.0 = 62.8 lbs — exact match. Adjustment internally consistent.

VO2 Max: Feb 2026 Garmin estimate of 47.0 ml/kg/min (at 167.5 lbs). Absolute VO2 ~3,571 ml/min held as forward projection anchor. At May 18 weight (165.8 lbs) implied relative score = 47.5 ml/kg/min. **Trade-off to note:** because absolute VO2 is held flat while mass is added, *relative* VO2 is projected to fall 47.5 → 42.7 by 2030 (see table). That is a deliberate physique-priority trade — added muscle lowers ml/kg/min by construction. If preserving/raising relative VO2 becomes a priority, it requires either holding mass or genuinely raising absolute VO2 (more VO2-max + Zone 2 volume), which the current hypertrophy phase de-prioritizes. **Source correction (v10):** the 47.0 is a **Garmin wearable estimate** (Forerunner 235 + chest strap, same 8-yr device) — *not* a lab CPET. So the absolute anchor (~3,571 ml/min) is wearable-derived and carries the ±3-8% wearable error; the monthly Garmin trend is the *same* instrument, so it confirms a stable trend but is not an independent measurement. A one-time CPET would give a method-independent number; until then, treat the absolute anchor as approximate.

Bio age — eVolt 43; "fitness-age proxy" ~30 (de-emphasized in v10). The ~30 figure is a *bespoke, unvalidated* composite (VO2 max 0.70, grip 0.15, cardiometabolic 0.15), each axis mapped to an age-equivalent and blended — not an Attia-published score and not a biological-age clock. It says "fitness profile of a man roughly 15-20 yrs younger," which the lifelong aerobic base (running since 1989, Ironman 1996/1998, Boston 2000, PB 3:09) supports — but it is **not a longevity measurement** and is treated as directional only. Note the VO2 anchor (47 at 50) maps to an above-average *untrained* 30-yr-old, not a trained one (55-60+). Real bio-age work (epigenetic clocks) and the high-evidence longevity markers are not in this dataset — see Appendix. Longevity is intentionally secondary here; physique and performance are the goals.

May 12 bloodwork story: Predictable response to two protocol changes on Apr 18-19 — stopping Anastrozole and raising T from 80→100mg/wk. Both pushed E2 upward; both landed where expected.

E2 194 → Anastrozole restarted May 13: Dr Salim resumed Anastrozole at exactly half the prior dose — 0.25mg × 2/wk (0.5mg/wk total). Matches the projected response from the May 12 alert. ~3-4 wks to steady state. Next draw is the true read.

Hematocrit + BIA share a root cause: Hct 0.522 → 0.498 with Hgb flat at 171 — plasma volume expanded. Same mechanism inflates BIA muscle/lean reads slightly. Two visible artifacts, one underlying mechanism — both small in magnitude given the small ECF differential.

TSH artifact confirmed: Free T4/T3 both mid-range normal. The TSH elevation is the Sermorelin/GH-axis fingerprint Dr Salim described in March. No action.

Metabolic wins: HbA1c 5.6→5.4 (below at-risk threshold), glucose 5.5→5.2, CRP <0.5. Reflects training volume + improving body composition. Not water-artifact sensitive.

ATHLETIC TRAINING CONTEXT — FOUNDATIONAL

LIFELONG TRAINING SUBSTRATE · CONTEXT FOR INTERPRETING ALL BODY COMPOSITION AND BIO AGE METRICS

The training history that contextualizes everything

Running: consistent since 1989 — 37 years. Not casual. Race résumé includes Ironman 1996, Ironman 1998, Boston Marathon 2000 (qualifying time required), marathon PB 3:09 in 1999 (~7:14/mile pace · top 5-10% of marathoners), half marathon 1:29:39 in 2022 at age 47 (6:50/mile pace · top decile masters). This is a lifetime endurance athlete profile.

Bouldering: consistent 2× per week since 2017 — 9 years. Sustained progressive overload on forearm flexors, finger flexors, grip endurance, and upper-body pulling strength. Explains the elevated grip strength reading and the connective tissue durability supporting the current heavy lifting block.

Heavy compound lifting: serious since Nov 28 2025 — 24 weeks. This is the only "beginner window" in the broader training picture. The prior sporadic lifting (Nov 2024-Nov 2025) layered onto the climbing-trained upper body, but structured heavy compound work is new.

How this reframes the metrics on this dashboard

VO2 max 47.0 ml/kg/min at age 51 is not impressive on its own — it's the expected output of 37 years of consistent aerobic training. Lifelong endurance athletes typically retain 70-80% of peak VO2 max into their 50s. Your 2022 baseline of 53-54 at 148 lbs likely reflects a peak in the late 20s / early 30s of ~60+ ml/kg/min. The "essentially unchanged absolute VO2 of ~3,571 ml/min in 3.5 years" isn't luck — it's the signature of someone maintaining cardiac stroke volume, mitochondrial density, and capillary infrastructure built decades ago.

Grip strength is partly climbing-specific. The ~49 kg Jamar-equivalent reading isn't a globally elevated strength signal — it's a trained-specific output. Recreational climbers with 5+ years of consistent training typically show grip 20-30% above untrained peers of similar muscle mass. The reading is still a positive longevity signal, but more confirmatory than independent.

The "beginner window" muscle gain framing needs adjustment. You are a neuromuscular beginner at heavy compound lifting (true) but not a beginner athlete (decisively not). Connective tissue, recovery capacity, motor unit recruitment, metabolic flexibility — all benefit from decades of training. The +13 lbs muscle since Aug 2024 is impressive given a slow-twitch-dominant, endurance-trained substrate. Pure novice gains might come faster early but plateau sooner — your trajectory is more sustainable.

Phase progression cautions are about concurrent training interference, not aerobic tolerance. The reason for the 6-cycle minimum in Phase 1 is the mTOR/AMPK interference effect[1], not "establishing tolerance" for structured run training. Distinction matters for how aggressively progression can proceed.

What this does to the fitness-age proxy

The composite lands near **~30 (range 28-33)** as a **directional fitness-age proxy** — read as "the fitness of a man roughly 15-20 years younger," not as a literal biological age. The training-age substrate unifies the three measured axes: a 50-year-old with 37 years of consistent aerobic training, 2 Ironman finishes, a sub-3:10 marathon PB, a 1:29 half at 47, and (per Garmin) improving HR-at-pace efficiency over 3 years has a physiological base no body-comp algorithm captures. **But the number is a bespoke, unvalidated construct, not a longevity measurement**; the VO2 anchor (47 at 50) maps to an above-average *untrained* 30-year-old, not a trained one. Treat it as a motivating directional signal and nothing more — the actual longevity markers worth tracking are in the Appendix, not here.

Garmin- and race-corroborated validation

The 37-year endurance substrate is independently corroborated by 3 years of Garmin data (630 runs, 3,597 km, May 2023 → May 2026), wearable VO2 max trends, and race PDFs from 2021-2022. The clearest signal is a **15-20 bpm reduction in heart rate at the same training pace** across the dataset — aerobic efficiency has substantially improved despite carrying ~18 lbs more body mass than in the 2022 half-marathon-trained state. Garmin VO2 max held 46-49 (avg 47.6) over the past 12 months; the Feb 2026 47.0 figure is itself a Garmin reading (same instrument), so this is a stable single-method trend, not independent corroboration. The race history corroborates the *engine*, not the precise number. The 2022 half marathon (1:29:38 at age 46/47, 4:14/km, disciplined HR-controlled execution) and 2021 5k pace (~4:01/km, max HR 191 at age 46) anchor the race-pace history. **Full synthesis in the Running History section below.**

Forward implications

Projection model: 2026-2030 muscle gain targets remain reasonable but should be framed as trained-athlete gains on a 37-year aerobic + 9-year climbing substrate — not as beginner gains. Magnitudes hold; narrative changes.

Phase progression: The conservative Phase 1 → Phase 2 timeline (6+ cycles minimum) remains appropriate for managing concurrent training interference, not because aerobic conditioning needs establishing. Once green lights hit, advancement should be confident.

Climbing as ongoing input: The 2× per week bouldering should be tracked alongside the lifting and running cycles. It's a meaningful weekly volume contribution and affects recovery debt calculations. Currently not represented in the 6-day rolling cycle structure.

Endnote [1] — mTOR/AMPK interference effect (mechanism)

The mTOR/AMPK interference effect is the molecular basis for why endurance and resistance training compete when stacked too close together on the same muscle group. The mechanism, in your terms:

mTOR (mechanistic Target of Rapamycin) is the master anabolic switch. Heavy resistance work — high mechanical tension, especially eccentric loading — activates the PI3K/Akt/mTOR cascade, which drives muscle protein synthesis. This is the signal that says "build."

AMPK (AMP-activated protein kinase) is the cellular energy sensor. When ATP is depleted and the AMP:ATP ratio climbs — which is exactly what sustained endurance work does — AMPK activates to restore energy balance: it upregulates mitochondrial biogenesis (via PGC-1α), fatty acid oxidation, and glucose uptake. This is the aerobic adaptation signal.

The conflict: AMPK directly inhibits mTOR. It phosphorylates TSC2 and Raptor, suppressing the mTORC1 complex. So when you finish an interval session and AMPK is elevated, then try to drive a hypertrophy stimulus, the anabolic signal is being actively damped at the molecular level. The cell is in "conserve energy" mode, which is antagonistic to "spend energy building tissue."

Three things specific to your situation that matter:

1. **It's largely a local, same-muscle-group phenomenon.** This is why your upper-body-weights-then-intervals combination works — the AMPK elevation from leg-dominant running doesn't meaningfully suppress mTOR signaling in trained upper-body tissue. It's also precisely why heavy legs + a hard run on the same day is prohibited: you'd be raising leg AMPK right when you want leg mTOR maximal.
2. **Your TRT + Sermorelin stack partially buffers it but doesn't eliminate it.** Elevated testosterone and the IGF-1 from the GH axis both feed into Akt/mTOR upstream, raising the baseline anabolic drive against which AMPK is fighting. That's a tilt in your favor, not a cancellation — the interference window is still real.
3. **The 6-cycle minimum in Phase 1 is an interference-management decision, not an aerobic-tolerance one.** That distinction is in your dashboard notes for a reason: you have ~37 years of aerobic base, so "tolerating" structured running was never the question. The conservative ramp exists to protect the leg hypertrophy signal during the muscle-building priority phase, since you're trying to add the interval stimulus without paying for it in blunted lower-body gains.

Practical upshot: the sequencing architecture you already follow — temporal separation of the competing signals (never legs + hard run same day, never run on Day 3 or 4) is the cleanest lever, because the interference is acute and time-dependent. Separate the signals in time and you get both adaptations; stack them and AMPK wins the local argument.

Annual training volume (Garmin)

YEAR	RUNS	DISTANCE	HRS	AVG HR	WTD PACE	BEST SINGLE PACE	CONTEXT
2023 (May-Dec)	164	989 km	99	148	5:59	5:18 (HM-distance Austria)	Pre-TRT base · ~1,580 km annualized
2024	223	1,463 km	146	153	5:59	5:10 (10.3km Tokyo)	TRT started Aug · highest volume year
2025	171	844 km	84	135	6:00	5:26	Volume halved · HR-at-pace drops sharply
2026 (Jan-Jul 21)	96	383 km	41	135	6:28	5:26	~679 km annualized · illness + hamstring disruption (see Interruption Log)

Wtd Pace = total time ÷ total km. Best single pace = fastest single-run weighted pace ≥3.5 km. "Avg HR" is the unweighted mean of per-run averages.

Revised (v11): HR-at-pace fell — but the magnitude is partly a run-duration artifact

Correction to the v10 framing. v10 called this "HR-at-pace dropped 15-20 bpm" and read it as a clean aerobic-efficiency gain. Re-examination on the fuller dataset says the effect is real but smaller than stated. The confound: **median run duration nearly halved over the same window** — ~44 min in 2024 vs ~29 min in 2026 — and cardiac drift alone raises whole-run average HR on longer efforts by roughly 5-8 bpm at identical pace. A meaningful slice of the "improvement" is therefore comparing long runs to short runs, not fitter to less fit. **The tighter test** — comparing periods of similar run length — shows median avg HR at ~6:00/km going *up* from 134 (2025 H1) to 139 (2026 YTD). Read the engine as **maintained**, not clearly improving. That is still a good outcome while adding ~18 lbs of mass, but the earlier framing overstated it.

The underlying series, presented as-is:

- **2024 H1 (pre-TRT trained baseline):** ~158 bpm average HR at ~6:00/km
- **2024 Q4 (TRT 2 mo · Anastrozole adjustment):** ~140 bpm at ~5:51/km
- **2025 H1 → 2026:** stable at **~135-138 bpm at ~6:00/km**

Net of the duration confound, the defensible read is a **real but moderate** improvement in aerobic efficiency across the 24-month window, achieved while carrying ~18 lbs more body mass than the 2022 half-marathon-trained reference state. Mechanisms consistent with a shift of this size: increased cardiac stroke volume, higher mitochondrial density in Type I fibers, improved capillary density, lactate threshold shifting right. Some of this is likely the protocol (TRT + Sermorelin), some is the more structured training pattern replacing earlier ad-hoc running. **The durable claim is that the underlying engine has not regressed while mass was added — which is the outcome the concurrent-training design was built to protect.** Claims beyond that are not supported by this dataset.

One further caveat surfaced in v11: habitual pace has been effectively frozen at ~6:00/km for three years (5.95 · 6.01 · 5.96 · 6.00 · 6.04 across successive periods). Weighted 2026 pace of 6:28 reflects deliberate load management during the injury window, not decline. But the flat multi-year pace means there has been **no meaningful speed exposure between 6:00/km and Helgerud rep pace (~4:00-4:30/km)** — a gap with direct injury implications (see Interruption Log below).

Garmin wearable VO2 max trend (Jun 2025 → Jul 2026) — extended in v11

14 monthly readings: **48, 48, 49, 48, 48, 48, 46, 47, 47, 47, 47, 48, 48, 47** (average **47.6**). The Dec 2025 dip to 46 lines up exactly with onset of serious lifting (Nov 28) and Sermorelin (Dec 22) — expected concurrent-training interference signal and protocol adaptation. Recovery to 47-48 followed, with **48 held through May and June 2026** and a return to **47 in July** — coinciding with the illness and hamstring disruption window.

> **Projection-anchor decision (v11): the 3,571 ml/min absolute anchor is retained unchanged.** The May-June 2026 reading of 48 briefly implied ~3,610 ml/min at 75.2 kg, and an interim recommendation was made to raise the anchor. **The July reading of 47 withdraws that recommendation** — 48 was a two-month excursion inside normal wearable variance, not a new level. Holding 3,571 is the conservative and correct call. Logged explicitly because the reverse error — ratcheting an anchor up on two favourable months — is exactly the failure mode this dashboard is built to avoid.

Read across the disruption: eight weeks containing three colds, four hamstring re-tweaks, and multiple weeks of overseas travel cost **one point** of wearable VO2 max. Detraining literature puts losses at roughly 5-10% after four weeks of *complete* cessation; running never stopped entirely here, and a 37-year aerobic base is highly resistant to short interruptions. This is the goal hierarchy working as designed — performance held while the primary physique objective continued uninterrupted.

Garmin VO2 max is wearable-derived and runs ±3-8% off lab values, but the **trend is the signal**, and the trend is stable-to-slightly-rising. The Feb 2026 47.0 figure is itself a Garmin reading from this same instrument — so the trend is internally consistent on a stable 8-yr device (reliable as a *trend*), not a cross-method validation. A CPET would be needed for a method-independent number.

Race-pace evidence from 2021–2022 (pre-Garmin)

- **Oct 17 2021 · 5 km in ~19:53 at age 46.** 5.13 km recorded in 20:36 (~4:01/km pace · 5.00 km projects to 19:53). Avg cadence 88, max HR 191, headwind 23 km/h, 10°C overcast. With HR_max now confirmed at 186 at age 51, this 191 at age 46 reads as a genuine near-max reading consistent with age-related decline (~1 bpm/yr) — not the sensor artifact it was previously flagged as.
- **Mar 5 2022 · half marathon 1:29:38 at age 46 (turned 47 May 30).** 21.3 km, 4:14/km, avg HR 168, max HR 184, cadence 90 throughout, 150 lbs morning weight, Vaporfly Next% shoes. Lap splits 4:13 / 4:16. Race notes: HR kept ≤160 first half, opened in second half, eased back at 15-18 km to ensure finish, pressed last 1 km. **Fastest half since 1999 at age 24.** The pacing discipline matters as much as the time — this was a controlled performance with margin, not a maximum effort.

Reconciling 2022 race form with current state

The 2022 half marathon was on a 148-150 lb / ~12% BF body trained specifically for distance running. Current body is 165.8 lbs / ~12.7% BF — **~18 lbs heavier, most of it muscle**. Despite that added mass, the HR-at-pace data shows the cardiopulmonary system is at least as efficient now as it was then, possibly more so. The simple arithmetic: if you could hold 4:14/km at 148 lbs in 2022, and your current absolute VO2 (~3,571 ml/min) is unchanged, then at 148 lbs again the relative VO2 would return to ~53 ml/kg/min — roughly the 2022 level. **A sub-1:25 half marathon is mathematically reachable at 148 lbs with current efficiency.** That is a forward-looking statement about potential, not a near-term target — the current cycle isn't structured to chase race performance, and the muscle-building phase is the strategic priority.

Recent training pattern (Feb 19 → May 11 2026, 12 weeks · Phase 1 execution check)

72 runs · 301 km in the Phase 1 window. Distance profile concentrated in the 3.5-6.5 km range, exactly what the 6-day rolling cycle prescribes. Several sessions show the Helgerud signature (max HR 170-176, avg HR 145-151 over 4-6 km) and tempo-run signature (max HR 165-170, avg HR 138-146, sustained 5:30-5:50/km pace). **Recovery runs are appropriately easy** — avg HR 110-130 at 6:30-7:30/km, indicating discipline on the easy days. The data does not yet show a clear pattern of degraded interval quality across cycles, which is a Phase 1 → Phase 2 green light to track explicitly. Longest run in the Garmin window: 12.15 km (vacation Austria 2023). No HM-distance test runs in the current cycle — appropriate for the muscle-building phase.

Forward implications for the training plan

HR_max confirmed at 186 bpm — Garmin Forerunner 235 + Garmin chest-strap HRM (per Ken · June 2026). Source matters: a chest strap (not optical wrist) is the reliable consumer HR source, and it's the same device Ken has run for 8 years — no cross-device calibration drift. The figure reconciles with the prior signals rather than contradicting them: the 2021 race PDF showed 191 at age 46 (also chest-strap), the lone Garmin reading of 189 (Feb 2026) sits within 3 bpm, and 186 at age 51 fits the typical ~1 bpm/year age decline. Three device-consistent readings clustering 186-191 is the corroboration that was missing when 191 stood alone. *Caveat:* this is an observed training peak — a firm lower bound, not a formal maximal test; if the absolute ceiling ever matters, a max-effort or clinical test settles it. The earlier 170 working estimate was low by ~16 bpm.

Zones re-anchored to 186. Recomputed targets: **Helgerud 90-95% = 167-177 bpm**; **tempo 80-83% (low Zone 4) = 149-154 bpm**; Zone 2 (60-70%) = 112-130 bpm. RPE remains the cross-check — Helgerud reps should feel barely sustainable (single-word answers), tempo uncomfortable but short-sentence conversational. Where Garmin HR and perceived effort disagree, trust perceived effort.

Phase 1 intervals were on-target all along. The observed interval ceiling of 170-176 bpm equals 91-95% of 186 — squarely inside the Helgerud 90-95% band, not "above target" as the old 170 anchor implied. So the interval-quality signal for Phase 2 readiness is cleaner than it looked: the prescribed intensity was being hit. One adjustment the new anchor surfaces — the tempo block (sustained portion, not whole-run average) should sit at **149-154 bpm**; the previously observed tempo max of 165-170 was running hot for a true tempo. Green-light criteria from Section 7.1 (sustained quality across reps, legs holding steady, recovered by Day 6) plus the 6-cycle floor remain the gate; the HR-zone question is now settled rather than approximate.

Climbing is a hidden volume input. 9 years of 2× per week bouldering doesn't appear in the Garmin run data but contributes meaningfully to upper-body and grip training load. Should be tracked alongside the 6-day cycle for honest recovery accounting.

LIFE CONTEXT & CONFOUNDERS — NEW IN V12

⚑ REVISED TIMING (V14) · THE INFLECTION IS MID-MAY, NOT LATE MARCH

Athlete correction, Aug 2026: sleep quality declined sharply around May 15, not in late March, and the strength plateau began at the same time. Three independent lines now converge on that date:

- **Self-reported sleep decline** — ~May 15
- **Self-reported strength plateau** — ~May 15; no working-weight increase since
- **Monthly run volume** — Mar 14, Apr 16, **May 11**, Jun 12, Jul 13, Aug 8 (partial, on pace ~16). The April peak of 16 was never reached again. This is objective and device-derived.

Everything adverse in the 2026 file sits after mid-May: IGF-1 183 (May 12) → 134 (Jul 28), the three colds (May 12, Jun 4, Jul 22), the four hamstring tweaks (from Jun 19), and the fat gain between the May and August scans.

Hypotheses ruled out: (a) *Low estradiol.* E2 was under 40 pmol/L from early 2025 through Feb 2026 — including the productive lifting period from Nov 28 2025. Current 113 is roughly three times the level at which gains were made, so low E2 cannot explain the plateau. Anastrozole also takes 3-4 weeks to reach steady state, so a May 13 restart cannot produce a May 15 effect. (b) *Late-March company founding as sole cause.* Six weeks too early for the strength stall.

Confounded week: May 12-15 contains the first cold of the year (May 12-23), the anastrozole restart (May 13), peak recorded E2 of 194 (May 12), and the reported sleep decline (~May 15). The illness alone is a plausible trigger for a plateau that then never regained momentum.

Measurement gap: the watch is not worn consistently, so Garmin resting HR and sleep data are both unusable. Manual bedtime/wake logging is the available instrument.

⚑ STRUCTURAL · THREE COMPANIES FOUNDED FROM LATE MARCH 2026 (TBG · PROVA · HCHW)

AI-first ventures. **Stress elevated, sleep compromised — self-reported.** This is the single largest uncontrolled variable in the 2026 dataset and was invisible until Aug 2026. It post-dates the Feb 24 draw and precedes every subsequent adverse signal:

- **IGF-1 decline** (214 → 183 → 134) — GH secretion is slow-wave-sleep dependent
- **Three colds in 11 weeks** (May 12-23, Jun 4-10, Jul 22-27) — none in the record before late March; sleep restriction is a strong predictor of URI susceptibility
- **Four hamstring tweaks** (Jun 19, Jul 5, Jul 11, Jul 19) — short sleep is linked to elevated soft-tissue injury rates; compounds the existing speed-exposure gap
- **CRP 2.5** — sleep restriction raises CRP/IL-6 independently of the cold
- **HbA1c/glucose drift** — sleep restriction impairs insulin sensitivity
- **End-2026 lean-mass target** — sleep restriction blunts muscle protein synthesis. Direct headwind on the PRIMARY goal.

Ken is addressing the sleep schedule as of Aug 2026. Log bedtime/wake for the 14 days preceding the next draw so the panel is interpretable.

◇ OPEN · SLEEP APNEA SCREENING — DECLINED FOR NOW (AUG 5 2026)

Raised because nocturnal hypoxia drives EPO → erythrocytosis, and testosterone therapy can worsen apnea — a recognised loop in TRT management. Would change the *cause* of the Hct trend rather than just the monitoring. **Ken elected to skip; Dr Salim did not respond to the question.** Caveat on record: apnea produces fragmented, hypoxic sleep at full duration, so fixing bedtime schedule would not resolve it — a null result from the sleep fix should not be read as excluding apnea. Cheap intermediate step: ask partner about snoring, gasping, or witnessed pauses. Not closed permanently.

SUPPLEMENT & NON-PRESCRIBED INTAKE LOG — NEW IN V12

ITEM	DOSE	STATUS	NOTE
Vitamin D3	2,000 IU/day	Starting Aug 2026	Dr Salim-specified dose. With fat-containing meal. Retest 3 months.
Vitamin K2 (MK-7)	100–200 mcg	Optional	Pairs with D3 given calcium at range floor. Evidence moderate, not strong.
Creatine monohydrate	5 g/day	Continue	Fully saturating at current mass. Inflates serum creatinine — cystatin C requested.
Ashwagandha	dose TBC	Washout before draws	**Undisclosed until Aug 2026.** Raises testosterone on bloodwork (Dr Salim, Jul 30). Also documented thyroid effects — confounds the TSH trend. Rare hepatotoxicity reports; ALT 22, no signal. Stop several days pre-draw.
Vitamin C	dose TBC	STOPPED Aug 2026	**Undisclosed until Aug 2026.** Markedly increases dietary iron absorption — a tailwind on the Hct problem, and weakens the iron-restriction read. Also blunts hypertrophy/mitochondrial adaptation at high dose around training.
Iron — any form	—	CONTRAINDICATED	Check all multivitamins, greens powders, endurance-marketed protein blends.
"Estrogen support" (DIM, CDG, chrysin, nettle root, high-dose zinc)	—	AVOID	Aromatase-inhibiting; stacks on prescribed Anastrozole. E2 113 is on target.
Boron / "T-boosters"	—	AVOID	Free T 784 vs ceiling 475. More androgen drives the red-cell problem.

Doses marked TBC must be filled in. *Three separate v11-era analyses (TSH resolution, iron-restriction read, free-T interpretation) were built on an incomplete picture of intake. This table exists so that does not recur — it captures everything taken, not only what Dr Salim prescribes.*

BLOOD PRESSURE — NEW SERIES IN V12

◇ FIRST READING EVER · 126/81 · PULSE 82 — NOT A BASELINE

PharmaSmart kiosk, GoodLife, Jul 30 2026. **Taken immediately after 60 min heavy resistance training — uninterpretable for baseline purposes.** Post-exercise hypotension would push the true resting value *higher*; residual post-set elevation would push it lower. The two effects run opposite and timing determines which dominates. Pulse 82 is post-exercise tachycardia and carries no fitness information.

By Hypertension Canada criteria (threshold 140/90) this is normal; by ACC/AHA 2017 (diastolic cutoff 80) it would read Stage 1. Not a problem — but diastolic is the number to watch given testosterone modestly raises BP and Hct 0.521 raises viscosity.

Action: validated home monitor (BIOS Diagnostics, Hypertension Canada-recommended, cuff 24–43 cm). Measure arm circumference first. Protocol: 2× morning pre-training, 2× evening ≥2 hrs post-session, 7 days, discard day 1, average the rest. Home threshold is **135/85**, not 140/90. That average is what Dr Salim can act on.

Also worth pulling: Garmin resting HR trend Jan → Aug 2026. If RHR drifted up from late March, that is objective longitudinal corroboration of the sleep/stress story from an 8-year continuous device.

RUN RESTART & BURST SESSION LOG — NEW IN V14

✓ RUNNING RESUMED · STAGE 0 CLEARED · NOW OPERATING AT STAGE 2 EQUIVALENT

Athlete report (Aug 15 2026): back to 3-4 runs/week, with 30-second bursts every 3 minutes, 6 reps, at least once weekly. Leg work also restarted — no longer at zero. The dashboard carried "Stage 0 · no running, eccentrics only" from late July until this update; that was stale. The burst structure is the correct intervention for the documented *speed-exposure gap* — habitual pace frozen near 6:00/km for three years against a Helgerud rep pace of 4:00-4:30/km with nothing in between.

DATE	DISTANCE	TIME	AVG PACE	AVG HR	MAX HR	% HRMAX	BEST PACE	READ
Aug 1 2026	3.58 km	22:49	6:23/km	127	148	80%	4:18/km	Clean — below injury band
Aug 4 2026	4.16 km	26:18	6:20/km	129	157	84%	4:05/km	Clean — below injury band
Aug 8 2026	3.55 km	22:40	6:24/km	124	140	75%	5:37/km	Bursts absent or very soft
Aug 9 2026 Georgina — different course	4.02 km	25:27	6:20/km	126	166	**89%**	**3:13/km**	**INSIDE the injury band.** Best pace far faster than the other three — a sprint, not a burst. HR climbed late in the session rather than spiking evenly, suggesting reps hardened as the run progressed.

Injury band = 163-178 bpm (88-96% HRmax). *All four hamstring tweaks (Jun 19, Jul 5, Jul 11, Jul 19) occurred on runs reaching this range; every run held below ~140 bpm max was clean. Three of four August sessions sit safely below it.* ***Prescription: cap bursts by feel near the 4:00-4:20/km of Aug 1 and Aug 4; do not chase pace.*** *The speed-exposure bridge does not require 3:13/km, and the added risk is real on compromised sleep with four tweaks in the prior two months.* ***Avg HR is consistently 124-129*** *— squarely in the Zone 2 band (112-130), confirming the easy-baseline-plus-brief-spike structure is working as intended.*

◇ INSTRUMENTATION · SET THE WATCH TO RUN STRUCTURED INTERVALS

Bursts are currently done by feel, which is why Aug 8 shows no meaningful spike and Aug 9 drifted to 3:13/km. Programming the Forerunner with an explicit 30s hard / 3:00 easy × 6 workout would auto-split the laps, make the Laps tab directly readable, and — most importantly — stop reps hardening across the session, which is the specific failure mode in the injury history. **Going forward: interval sessions reviewed via the Laps tab (per-rep HR and pace); easy runs summarised by monthly count and volume only.**

TRAINING INTERRUPTION LOG & RETURN-TO-RUN PROTOCOL — UPDATED V14

WHY THIS SECTION EXISTS

Prior versions carried no record of illness, injury, or travel. The consequence was that the Garmin data read as *inconsistency* when it was in fact *disruption* — a materially different diagnosis with a different remedy. Any future review of the May-July 2026 run record must be read against this log, not against the prescribed 6-day cycle in isolation.

The log

DATES	EVENT	TRAINING IMPACT	GARMIN CORROBORATION
May 12 – 23	Cold (1 of 3)	Sessions lost	6-day gap May 11 → May 18
Jun 4 – 10	Cold (2 of 3)	Sessions lost	Volume held but intensity low
Jun 19	**Hamstring tweak 1** · muscle belly	Several days rest · Helgerud stopped	5.20 km · avg 126 · **max 164 (88%)**
Jul 5	**Hamstring tweak 2**	Several days rest	5.69 km · avg 151 · **max 178 (96%)** · TE 4.4 — hardest session of the window
Jul 11	**Hamstring tweak 3**	Several days rest	5.28 km · avg 133 · **max 166 (89%)** · preceded by aborted 0.50 km run same morning
Jul 19	**Hamstring tweak 4**	Several days rest · load management begins	3.58 km · avg 134 · **max 163 (88%)**
Jul 22 – 27	Cold (3 of 3) — ongoing at v11 date	Return-to-run start deferred	Last recorded session Jul 21
Interspersed	Overseas travel · several weeks	Normal cycle disrupted	Treadmill substitutions Jun 26, Jun 28, Jul 21

% figures are of confirmed HR_max 186. Athlete report: each tweak was caught at "100% → 95%", the session terminated, and the return jogged. Managing the acute event correctly is why the record shows four niggles rather than one grade-II tear.

Correction: the Helgerud record is not a compliance failure

v10 implicitly benchmarked the run data against a Phase 1 cycle assumed to be running since early 2026. That benchmark was wrong. **The Helgerud protocol was only learned in April 2026.** The four executed sessions — **Apr 11, Apr 15, Apr 22, Apr 28** (identifiable in the Garmin export by their 19–21 lap structure) — are therefore the *complete* history of the intervention, delivered in the three weeks between learning it and the disruption window opening. Execution quality on those four was good: peak HR 170–176, or 91–95% of HR_max, squarely in the prescribed band. **The problem was never adherence. It was that the block had barely begun before illness and injury closed it.**

Consequence for phase tracking: the Phase 1 six-cycle clock has *not started*. Green lights cannot be assessed against four sessions delivered inside a three-week window. Phase 1 begins when the return-to-run progression below completes.

Pattern finding: every tweak occurred on a run that crossed ~163 bpm

All four re-injury sessions reached **163–178 bpm (88–96% of HR_max)**. Not one occurred on a run held below that. The contrast is sharp against the deliberately controlled July sessions — Jul 4, Jul 6, Jul 9 — which peaked at only **127–134 bpm (68–72%)** and produced no symptoms. Median avg-to-max HR spread on tweak days was **31 bpm** versus **20 bpm** on clean runs.

The mechanism this points to is **uncontrolled surges inside nominally easy runs**. The Jun 19 tweak is the instructive case: whole-run average HR was only 126 — genuinely easy by any reading — yet it still spiked to 164. The Puslinch route carries 19–22 m of ascent, and hamstrings fail at high stride velocity, not high average effort. An "easy run" that contains a hill surge is not an easy run for this tissue.

> **Practical guardrail:** during return-to-run, cap *maximum* HR at **~150 bpm (80% of 186)**, not average HR. Walk the hills. Average HR is the wrong control variable here because it conceals exactly the surges that appear to be doing the damage — and max HR is already on the watch face, so this costs nothing to implement. **Caveat, stated plainly:** n = 4 events, observational, and elevated max HR may be a *marker* of terrain or surging rather than the cause. It is a defensible guardrail, not an established mechanism.

Root-cause gap 1: no eccentric hamstring loading at long muscle length

Current lower-body eccentric work is **controlled squats at ~50% of maximum**. That is real eccentric loading, but it does not load the tissue that is failing. Back squats are quad- and glute-dominant — hamstring activation sits at roughly 10–25% of MVIC, and critically the hamstring remains *short* throughout the movement. The failure mode in running is the opposite position: hip flexed, knee extending, hamstring lengthening under tension while decelerating the shin through terminal swing. **Squats never place the muscle there.**

Exercises that do, in ascending order of demand: **Romanian deadlifts** (hip hinge, long length, controlled eccentric) → **single-leg RDL** (adds pelvic control) → **Nordic hamstring curls** (begin band-assisted or as partials; unassisted Nordics are punishing for the untrained). The Nordic curl carries among the strongest injury-prevention evidence in sports medicine, with meta-analyses reporting roughly halved strain rates.

Why this belongs on the dashboard and not only in a rehab plan: RDLs and Nordics are hypertrophy work for the posterior chain. Adding them to Day 3 serves the *primary physique objective* and the injury problem with the same sets. Interventions that serve both goals simultaneously are rare in this program — this is one.

Root-cause gap 2: three years with no speed exposure between 6:00/km and rep pace

Habitual pace has sat at ~6:00/km essentially unchanged since 2023. Helgerud reps demand roughly **4:00–4:30/km**. There is nothing in the training record between those two speeds. Hamstrings tolerate high stride velocity by being *progressively exposed* to it; going from three years of near-exclusive 6:00/km running to repeated four-minute efforts well faster asks a tissue to perform at a speed it has not trained at since 2022.

The bridge is strides: 4–6 × 15–20 seconds of controlled acceleration with full easy-jog recovery, twice weekly, appended to easy runs. Low total volume, negligible fatigue cost, and they are the missing rung on the ladder. Strides are also the readiness *test* — tolerating them cleanly is the evidence that eccentric capacity has rebuilt.

Return-to-run protocol · staged

STAGE	RUNNING	LOADING	GATE TO ADVANCE
Stage 0 ✓ CLEARED Aug 2026	None. Treadmill/bike optional if symptom-free.	RDLs + band-assisted Nordics 2×/wk	Cold fully resolved. **Do not start Stage 1 while still ill** — returning to load mid-infection is how tweak 3 became tweak 4.
Stage 1 wks 1–3	1-min efforts @ 60% · walk recovery · **max HR ≤150** · walk all hills	Eccentrics 2×/wk continue, progressively loaded	**Duration only — never speed.** Build continuous easy running before adding any pace.
Stage 2 **◄ CURRENT (Aug 2026)**	Continuous easy running + **strides 4–6 × 15–20s**, 2×/wk	Nordics progressing toward unassisted	Strides fully symptom-free — this is the capacity test
Stage 3	Tempo reintroduced (149–154 bpm sustained block)	Eccentrics now permanent on Day 3	**2 clean weeks of strides** before any interval work
Stage 4	Helgerud 4×4 resumes · **Phase 1 six-cycle clock starts here**	Full leg day + eccentrics	Green lights per §7.1 assessed from this point forward

*The 1-min/60% progression is a **rehab protocol, not a training protocol**. It must not drift into becoming the substitute for interval work — it trains nothing relevant to VO2 max. Its only job is restoring continuous running. Note also the standing rule from §6: 10×1 min is never a Helgerud substitute.*

Structural finding: legs receive half the weekly exposure of upper body

The 6-day cycle gives upper body **two** exposures (Days 1/2 and 4/5) and legs **one** (Day 3) — roughly 1.2 leg sessions per week. Lower body is approximately half of total skeletal muscle mass. The hypertrophy literature is reasonably consistent that two exposures per muscle group per week outperforms one at matched volume. **With physique as the primary objective, this is the most likely structural constraint on the End 2026 target of 86.8 lbs.**

The injury window makes this cheaper to fix than it would otherwise be: running is minimal through Stages 0–2 anyway, so a second lower-body exposure can be introduced with little concurrent-training interference cost. **The trade to decide deliberately:** if Day 5 becomes a second leg session, it is no longer available as the Phase 2 landing spot for the second Helgerud. That is a real conflict between the primary and secondary objectives and should be resolved explicitly rather than by default.

Positive signal worth preserving: the treadmill substitution

Three treadmill sessions appear in the window — **Jun 26, Jun 28, Jul 21**. The Jul 21 session, two days after the fourth tweak, ran **28 minutes at avg HR 145 / max 169, Aerobic TE 3.8** — a genuine aerobic stimulus delivered with the hamstring protected. Distance fields read ~0.1 km (foot-pod miscalibration, ignore them); duration and HR are the valid data. **This is the right instinct and should be formalized:** incline treadmill work at controlled speed is the highest-value aerobic modality available during Stages 0-1, because it removes the stride-velocity spike that appears to be the injury trigger while retaining the cardiac load that protects VO2 max.

For Dr Salim — timing overlap worth raising

> **Anastrozole was restarted May 13 2026** at 0.25mg × 2/wk, following the E2 rebound to 194 pmol/L. Estradiol would have begun falling over the following 4-6 weeks. **The first hamstring tweak occurred June 19** — approximately five weeks later, near the point of maximal E2 decline. Low estradiol in men associates with reduced tendon and connective-tissue quality and impaired collagen turnover. The overlap is most likely coincidental — travel, illness, a three-year speed-exposure gap, and absent eccentric loading are all sufficient explanations on their own — **but the datapoint should sit with the physician rather than be omitted.** Relevant to the pending E2 retest (target 70-150 pmol/L): if E2 has overshot downward, that is now a connective-tissue consideration as well as a joint-comfort and bone-density one. *Surfaced for discussion; not a recommendation to alter dosing.*

Net read on the disruption window

The primary objective was never interrupted. Upper body, bouldering, and the hormonal protocol continued throughout; the hamstring constrains running, not lifting. The physique trajectory toward 86.8 lbs is intact and unaffected by this window — which is precisely the resilience the settled goal hierarchy was designed to produce.

The cost was one point of wearable VO2 max (48 → 47) across eight weeks containing three illnesses, four re-injuries, and multiple weeks abroad. Against a 37-year aerobic base, that is a rounding error. **The genuine cost is time-to-Phase-1**, which is now pushed out by roughly 5-8 weeks. That is an acceptable price for entering the block with eccentric capacity built and a speed-exposure ladder in place, rather than re-entering it on the same tissue that has now failed four times.

SCAN COMMENTARY LOG

REVIEW OF FEB 19 2026 EVOLT · 18 WKS POST-OCT 15 SCAN

What's working — clearly

First eVolt-to-eVolt confirmation that the system is firing. Two scans 18 weeks apart — though on different machines (Oct at GoodLife #281 Eramosa, Feb at #242 Pergola). Cross-machine eVolt comparisons can carry ±0.5-1 kg lean variance, so the deltas warrant some interpretive caution. With that caveat: skeletal muscle +2.2 lbs (35.2 → 36.2 kg) is solid, particularly since this 18-week window included only the first ~12 weeks of serious training (started Nov 28 2025). The earlier 6 weeks of the Oct→Feb interval were still in the sporadic ~1×/wk training pattern. The muscle accrual is essentially all from the last 12 weeks.

Rate of ~0.18 lbs/wk over the serious-training window is strong for a 50-year-old. Even accounting for TRT support, this is at the upper end of what published literature shows for trained men in their 50s during early-phase hypertrophy work. Slow-twitch dominance was supposed to limit the ceiling — early evidence suggests the protocol overrides that constraint at least in the beginner window.

The Sermorelin layer is producing visible effects already. Started Dec 22 2025 — only ~8.5 weeks before this scan, and likely partial steady-state. Yet body composition is still trending favorably (BF% essentially flat at 14.3 → 14.6% despite a 5.5 lb weight gain, meaning fat:muscle ratio of the gain is roughly 1:1.7 — a respectable recomp signature). The Feb 24 bloodwork (drawn 5 days after this scan) will tell the IGF-1 story.

BMR up 41 kcal — small but in the right direction, confirms the lean mass gain is metabolically active tissue, not artifact.

What's working — but worth a caveat

+5.5 lbs weight in 18 weeks is on the aggressive side. The +2.2 lbs muscle / +1.3 lbs fat ratio is good (1.7:1 recomp), but the absolute weight delta is ~0.3 lbs/week — fast for a 50-year-old not in a deliberate bulking phase. Some of this is appropriate (glycogen + water + lean tissue accompanying serious training onset), but worth watching: if the next 18 weeks repeat this pace without a corresponding muscle:fat ratio, the trajectory tips from recomp toward overfeed.

TBW jumped from Optimal (45.4 kg) to High (46.7 kg) — first time it's been flagged. ECF rose +0.8 kg (15.1 → 15.9 kg). This is worth noting: the elevated water state means BIA muscle/lean reads here are probably modestly inflated (perhaps 0.3-0.5 lbs of the +2.2 muscle is water signal rather than tissue). The underlying tissue gain is real, just slightly less than headline.

Body fat % ticked up 14.3 → 14.6%. Tiny, but not the direction expected when Sermorelin is supposed to be driving fat oxidation. Plausible explanations: Sermorelin too new to show fat-loss effects yet, caloric intake increased to support training, or measurement variance well within BIA precision (~0.5%). Not a concern at this magnitude, but the Feb 24 bloodwork's IGF-1 reading will help distinguish "Sermorelin not yet engaged" from "Sermorelin engaged but caloric surplus dominating."

Bio age ticked from 44 → 45. Mildly concerning at face value — this is the direction nobody wants. But the composite is BF%-sensitive and the small BF% uptick likely drives it. If the next scan shows BF% trending down with continued muscle gain, bio age should respond.

Visceral fat area essentially unchanged (80 → 81 cm²). Not improving, not worsening. Consistent with the modest BF% tick — the body is in a build phase, not a leaning phase. Expect this to start trending down once Sermorelin reaches full effect and the training load stabilizes.

The Oct 15 → Feb 19 comparison includes the start of serious training mid-interval. Reading the rate as 0.12 lbs/wk muscle gain is technically correct over the 18-week window, but obscures the underlying truth that ~0 lbs was gained Oct→Nov 28 and ~2.2 lbs was gained in the 12 weeks since. The "real" hypertrophy rate to anchor projections is the 12-week serious-training rate, which is closer to ~0.18 lbs/wk.

Pace check against the projection

If the ~0.18 lbs/wk pace of the first 12 serious-training weeks held for the rest of 2026, that'd be roughly +8 lbs muscle over the next ~10.5 months — putting end-2026 muscle at ~88 lbs vs the current projection model's 86.8 lbs target. Modest beat. But beginner-window rates typically decelerate, so a ~+7.0 lbs gain Feb→Dec (the existing End 2026 projection) is a reasonable forward estimate. No revision needed.

VO2 max (Garmin wearable estimate) read 47.0 ml/kg/min — down modestly from the Aug 2022 baseline of 53-54, but that baseline was at 148 lbs / ~12% BF (post-half-marathon training). Absolute VO2 ~3,571 ml/min is essentially unchanged in 3.5 years — the relative score decline is entirely the weight gain, not a real cardiopulmonary regression. This is the projection anchor going forward.

The one thing actually worth watching

Bloodwork drawn Feb 24 will be the key follow-up. Three specific reads to look for: **IGF-1** (confirms Sermorelin response — if >200 ug/L at only 8.5 weeks in, that's strong pituitary response); **Hematocrit** (has been creeping up on TRT — May 2025 hit 0.529; if it's now >0.52, dose adjustment or phlebotomy enters the conversation); **Estradiol** (was <40 in May 2025 on Anastrozole — if still suppressed, dose may need to drop because over-suppression hurts joint health and bone density).

Bottom line

This is the first scan that lets us confirm the protocol is producing the predicted response in real time. The muscle gain rate during serious training is at the upper end of expectations for the demographic. The modest fat gain is the only mild concern, and likely resolves as Sermorelin reaches full steady-state effect and as caloric intake normalizes for the new training load.

Strategic implications: **no protocol changes needed**. Hold course on TRT/HCG/Anastrozole/Sermorelin. Maintain 6-day rolling training cycle. The Feb 24 bloodwork will determine whether any dose adjustments are warranted. The next eVolt (~May 2026) will be the read that tells us whether the 12-week beginner-rate hypertrophy is sustaining or normalizing.

Open question for next conversation: at what point is it appropriate to add the second weekly Helgerud interval (Phase 2 of the run protocol)? Currently in Phase 1; need 6+ cycles of clean Phase 1 quality before considering the progression. Probably not yet — recovery from the new training load is still settling in.

Recalibration addendum

Earlier in this session I applied a water-artifact correction that turned out to be too aggressive. The original correction (muscle −0.5 lbs, BF% +1.5 pts) was built on the assumption that May 18's elevated TBW was a step change from a "normal" Feb baseline. When full Feb 19 scan data became available, it turned out **Feb 19 TBW was also flagged High (46.7 kg)** — only +0.3 kg ECF below May 18. The differential between scans is small, so the correction should be small. Revised: muscle −0.2 lbs, BF% +1.3 pts. The numbers below reflect the recalibrated values. The original eVolt headline is closer to reality than the first correction suggested.

What's working — clearly

The fat loss is still the story, slightly smaller than headline. Adjusted figures show **−3.5 lbs fat in 13 weeks (not −5.5)**, with muscle continuing to build (+2.2 lbs adjusted) at flat-to-slightly-down weight. Clean recomposition without aggressive caloric restriction (BMR up, training hard, weight essentially flat). The Sermorelin signature shows up where you'd expect — fat oxidation while preserving lean tissue. The protocol is doing the work.

Visceral fat is the cleanest signal. Visceral fat level 9 → 7 (independent of BIA water effects), and visceral fat area 81 cm² → 59 cm² — a **27% reduction in visceral fat**. These are not water-artifact sensitive. Real adipose tissue came off, and the right kind: metabolically active fat around the organs.

The intracellular fluid story matters. ICF rose +1.0 kg from Feb to May (30.8 → 31.8 kg). ICF is water inside cells, primarily muscle cells. That is a real hypertrophy signal — cells are bigger, holding more water, doing more work. This is one reason the water correction should be small: most of the TBW rise was ICF (real), only 0.3 kg was ECF (artifact).

The metabolic panel is the quiet win. HbA1c 5.6 → 5.4 puts you back below the 5.5% at-risk threshold. Glucose 5.5 → 5.2. CRP holding under 0.5. These markers aren't water-artifact sensitive — they're the cardiometabolic profile that actually predicts the next decade.

What's working — but worth a caveat

The hematocrit "self-correction" is still partly a mirage. Hct 0.522 → 0.498 with Hgb flat at 171 — plasma volume expanded, red cell mass didn't decrease. Rising E2 was probably the dominant driver. With Anastrozole restarted, expect the plasma volume effect to partially reverse and Hct to drift back up over 4-6 weeks.

11.4% BF still reads slightly better than reality, but not as much as I previously claimed. Recalibrated to ~12.7% BF. The +1.3 pt correction is real but modest. Still excellent — and still a major improvement from Feb's 14.6%.

Muscle inflation is smaller than the prior correction suggested. Revised correction: 82.2 → 82.0 lbs (only −0.2 lbs). The Feb → May muscle gain is **+2.2 lbs (not the +1.9 I previously reported)** at rate ~0.17 lbs/wk — essentially matching the prior 0.18 lbs/wk eVolt-implied pace. The "decelerating" framing I used earlier was an artifact of my over-correction, not a real signal.

Bio age 43 (eVolt) vs ~30 (fitness-age proxy — de-emphasized). eVolt uses body composition only — no VO2 max, no grip, no bloodwork. The ~30 figure is a bespoke, unvalidated composite (VO2 0.70 / grip 0.15 / cardiometabolic 0.15); it is a directional fitness signal, not a longevity measurement, and is moved to the Appendix in v10.

Grip strength input (May 18 2026, Handeful digital dynamometer · best of attempts): Right hand 135.2 lbs / 61.3 kg · Left hand 129.8 lbs / 58.9 kg · R/L ratio 1.04 (excellent symmetry — typical is 1.10). **Caveat:** consumer digital dynamometers typically read 15-25% higher than Jamar hydraulic. Applying a 20% midpoint discount gives a true dominant-hand estimate of ~49 kg — **≥90th percentile for males age 51** (NIH Toolbox / JOSPT 2018), and roughly the 50th percentile for a 30-year-old male. Maps to grip-strength bio age of **~33 (range 30-35)**. Caveat: partly climbing-trained (9 years of 2x/wk bouldering).

Running-history corroboration (see Running History section above). 3 years of Garmin data show HR-at-pace has dropped 15-20 bpm at the same training pace — substantial aerobic-efficiency improvement despite carrying ~18 lbs more body mass than the 2022 half-marathon-trained state. Garmin monthly VO2 max averaged 47.6 over the past year; the Feb 2026 47.0 is the same Garmin instrument, so the two are one measurement system (consistent trend), not independent reads. Combined with the 2022 half marathon (1:29:38 at 4:14/km, age 46/47) and 2021 5K pace (~4:01/km, max HR 191 at age 46), the fitness-age proxy lands around **~30 (range 28-33) as a directional signal** — strong as a description of aerobic fitness, but not a longevity measurement (see Appendix).

Recommendation: get a Jamar reading at a clinic or PT office once to calibrate your Handeful's offset. After that, the Handeful is reliable for tracking trend.

Methodology & limitations — bio age: *The "Attia-weighted" figure is a* ***bespoke composite*** *that applies Attia's prioritization (VO2 max 0.70 · grip/strength 0.15 · cardiometabolic 0.15), each axis mapped to an age-equivalent then blended. It is* ***not*** *an Attia-published score and* ***not*** *a validated biological-age clock; the ~30 (range 28-33) is an interpretive estimate anchored to population norms, not a measurement.* ***Association ≠ causation:*** *the VO2 max ↔ all-cause-mortality evidence it leans on (Mandsager 2018, n=122,007; Kokkinos 2022, n>750,000, ~13-15% lower mortality per MET) is observational. Reverse causation, genetics, and survivorship/selection bias mean the causal payoff of training the metric is softer than the headline ~5× low-vs-elite gap implies — a limit Attia himself acknowledges.* ***Reliability:*** *even validated epigenetic clocks can swing several years on retest (meals, stress, circadian variation), so any single bio-age number — this one included — warrants humility.* ***Net:*** *the VO2-max-dominant prioritization is mainstream and defensible, not fringe; treat ~30 as a directional signal, not a precise figure.*

Pace check against the projection

End 2026 target is 86.8 lbs muscle. Adjusted current = 82.0. Need +4.8 over 7.5 months = ~0.6 lbs/month. Current adjusted pace ~0.7 lbs/month — on track or slightly ahead. The beginner window narrows from here; expect the next 13 weeks to deliver +1.5-2.0 lbs (adjusted) rather than +2.2. The body gets stingier as the easy adaptations get harvested.

BF% glide path (12.0 / 11.5 / 11.0 / 11.0 / 11.0%) is realistic on the recalibrated scale. Expect future eVolt headlines to read 0.5-1.5 pts lower than these targets when TBW is elevated.

The one thing actually worth watching

IGF-1 drift 214 → 183 (May 12 draw) is the only marker moving in an unhelpful direction. Still well in range, still well above pre-Sermorelin 161 baseline — not alarming yet. But the fat-loss engine on this protocol is largely IGF-1-mediated. If the next draw shows another step down toward 160, worth investigating before it becomes a trend. Causes range from benign (injection-to-draw timing) to worth-discussing (mild receptor downregulation, dose timing relative to peak). Raise at next Dr Salim conversation, not urgent.

Bottom line

24 weeks into serious training, four-pillar protocol stable, May 18 scan represents real progress — the headline numbers are mildly inflated by water effects but the recalibrated adjusted figures (+2.2 muscle / -3.5 fat / 14.6 → 12.7% BF) are very close to eVolt headline. Projection model holds at adjusted values. Action items are tactical (watch E2 settle, watch Hct rebound, monitor IGF-1 trajectory) — **no protocol pillar needs rethinking**.

The honest framing, now correctly calibrated: **+13.0 lbs muscle since baseline (adjusted)**, vs +13.2 headline. Earlier reported +12.7 was an over-correction. The system has been producing genuinely excellent results.

Running-side strategic question (updated June 2026): with HR_max now confirmed at 186, the zone-calibration uncertainty is resolved. The recent interval ceiling of 170-176 bpm reads as 91-95% of max — on-target Helgerud work, not over-reaching. Phase 2 readiness on the Section 7.1 green-light criteria looks close to triggered; deliberate review in the next cycle, with the 6-cycle minimum floor still applying.

APPENDIX · LONGEVITY WATCH-LIST & FOLLOW-UP TO-DO

WHY THIS IS AN APPENDIX (SECONDARY BY DESIGN)

Physique and performance are the dashboard's job. **Longevity is deliberately de-emphasized in v10** and parked here as a short tracking list, because the highest-evidence longevity levers are simple and mostly *absent from the current dataset* — they aren't the eVolt/VO2/hormone metrics the rest of the dashboard optimizes. These are **follow-up items to raise with Dr Salim and a primary-care physician**, not instructions. Adding a handful of them would do more for actual lifespan than any further refinement of the body-comp model. Evidence cited is observational; this section surfaces items, it does not prescribe.

HIGH-EVIDENCE MARKER	WHY IT MATTERS	STATUS	ACTION / CADENCE
ApoB	Causal driver of atherosclerotic CV disease — the thing most likely to actually end the projection. Outperforms LDL-C and CRP as a risk metric.	TO-DO · NOT MEASURED	Add to next LifeLabs draw. Discuss target with physician (primary-prevention reference often <0.8 g/L; lower if higher risk).
Lp(a)	Largely genetic, lifelong CV / aortic-stenosis risk. Measured once tells you how aggressive the rest of lipid control should be.	TO-DO · ONE-TIME	Order once at next draw. Result is stable for life — no need to repeat.
Blood Pressure	Single biggest *modifiable* driver of cardiovascular and dementia risk. Cheap, fast, currently a blind spot.	TO-DO · NOT TRACKED	Validated home cuff; seated AM/PM ~2×/wk; bring trailing averages to Dr Salim. (TRT can raise BP — worth a baseline.)
Sleep duration + regularity	Upstream of nearly everything this dashboard cares about — recovery, GH pulsatility, glucose, training adaptation. Highest-leverage free intervention.	TO-DO · NOT TRACKED	Use the Garmin/wearable sleep data already being collected; target 7-9 h with a consistent schedule; review monthly alongside training load.
CAC Score coronary artery calcium	A cheap one-time CT that anchors *actual* cardiac risk far better than any fitness-age proxy. A zero score is strongly reassuring; a non-zero score changes the plan.	TO-DO · DISCUSS	Raise appropriateness and timing with physician (commonly considered from ~age 40-50+).
Cancer Screening	The second major longevity pillar alongside cardiovascular. Especially salient given GH/IGF-1 use (see tensions below).	STATUS UNKNOWN TO DASHBOARD	Confirm colonoscopy done/scheduled (baseline from age 51), annual skin check; discuss any age-appropriate additions with physician.

Not medical advice. *Tracking and discussion items only — the dashboard surfaces them; Dr Salim and your physician decide and order. Cited evidence is observational/associational.*

PROTOCOL TENSIONS NOTED WITH EYES OPEN · PERFORMANCE/QOL CHOSEN OVER LONGEVITY, DELIBERATELY

IGF-1 direction (Sermorelin) runs against the longevity literature

The dashboard wants IGF-1 *higher* (the 214 → 183 drift is flagged as unwelcome) because IGF-1 mediates the recovery and fat-oxidation this protocol is built on. From a **longevity** angle that's backwards: lower GH/IGF-1 tone is what associates with exceptional human survival and lower cancer/diabetes risk (Milman 2014, *Aging Cell*; Junnila 2013, *Nat Rev Endocrinol*; Laron-syndrome cancer protection), and the rapamycin/CR interventions in that field work by *down*-regulating this exact pathway. **Magnitude check:** current IGF-1 (183-214) is physiologic and in-range, and Sermorelin's pulsatile GH is far gentler than exogenous HGH — so this is a modest, speculative trade, not an alarm. The point is only to file it correctly: pushing IGF-1 up is a **physique/recovery choice**, not an anti-aging one. Re-examine if priorities shift toward lifespan.

Muscle mass vs. relative VO2 max — a built-in conflict

Holding absolute VO2 flat (~3,571 ml/min) while adding mass means **relative VO2 falls by construction** — the projection table shows 47.5 → 42.7 ml/kg/min by 2030. Since VO2/kg is the single marker most tied to all-cause mortality, the muscle program and an aerobic-longevity goal pull in opposite numerical directions. Accepted here as a **physique-priority trade**. If relative VO2 ever becomes the priority, the levers are: hold mass, or genuinely raise absolute VO2 with more VO2-max + Zone 2 volume (the current hypertrophy phase de-prioritizes both).

Hematocrit — the one near-term hard-endpoint risk in the stack

This is the item that's an actual safety number, not a framing point. Hct hit 0.529, has lived 0.498-0.522, and the recent "drop" to 0.498 is most likely **plasma-volume dilution** (Hgb flat at 171) that may reverse now that T is at 100 mg/wk and Anastrozole is restarted — both push Hct up. Erythrocytosis drives hyperviscosity/thrombotic risk, and the large TRT trial (TRAVERSE 2023) showed small but real increases in atrial fibrillation, pulmonary embolism, and acute kidney injury; Hct is the modifiable piece. **Suggested follow-up:** agree a pre-set action threshold with Dr Salim (what Hct triggers a dose cut vs. therapeutic phlebotomy), and re-check Hct specifically against the 100 mg/wk increase on the next draw.

Supraphysiologic free T — QoL, not lifespan

Total T 32.7 and Free T 667 (≈40% above the lab ceiling) are a deliberate "higher but safe" target for mood, libido, and drive. Legitimate and Dr Salim's call — just with **no longevity rationale** and a small incremental contribution to the Hct/AF considerations above. Filed as a quality-of-life choice.

Bottom line for this appendix

Nothing here argues for stopping or shrinking the protocol — it's producing excellent physique and performance results, and the metabolic panel (HbA1c 5.4 / glucose 5.2 / CRP <0.5) is genuinely the best longevity news on the whole sheet. The asks are small and additive: **get ApoB + Lp(a) on the next draw, start a BP and sleep log, discuss a CAC score and screening cadence, and put a number on the Hct action threshold.** Those few items move the actual-lifespan needle more than any further model refinement — and they let physique and performance stay the headline without longevity going entirely untracked.